

Q1 2021 (01 Sep Shift 2)

All the arrangements, with or without meaning, of the word *FARMER* are written excluding any word that has two *R* appearing together. The arrangements are listed serially in the alphabetic order as in the English dictionary. Then the serial number of the word *FARMER* in this list is .

Q2 2021 (01 Sep Shift 2)

Let $P_1, P_2, \dots, P_{15}$ be 15 points on a circle. The

number of distinct triangles formed by points

P_i, P_j, P_k such that $i + j + k \neq 15$, is :

- (1) 12
- (2) 419
- (3) 443
- (4) 455

Q3 2021 (31 Aug Shift 2)

The number of 4-digit numbers which are neither multiple of 7 nor multiple of 3 is .

Q4 2021 (31 Aug Shift 1)

The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is .

Q5 2021 (27 Aug Shift 2)

Let $S = \{1, 2, 3, 4, 5, 6, 9\}$. Then the number of elements in the set $T = \{A \subseteq S : A \neq \phi \text{ and the sum of all the elements of } A \text{ is not a multiple of } 3\}$ is .

Q6 2021 (27 Aug Shift 1)

A number is called a palindrome if it reads the same backward as well as forward. For example 285582 is a six digit palindrome. The number of six digit palindromes, which are divisible by 55, is .

Q7 2021 (26 Aug Shift 2)

The sum of all 3 -digit numbers less than or equal to 500, that are formed without using the digit "1" and they all are multiple of 11, is .

Q8 2021 (26 Aug Shift 1)

The number of three-digit even numbers, formed by the digits 0, 1, 3, 4, 6, 7 if the repetition of digits is not allowed, is .

Q9 2021 (26 Aug Shift 1)

If ${}^1P_1 + 2 \cdot {}^2P_2 + 3 \cdot {}^3P_3 + \dots + 15 \cdot {}^{15}P_{15} = {}^4P_r - s, 0 \leq s \leq 1$

then ${}^{q+s}C_{r-s}$ is equal to .

Answer Key

Q1 (77)

Q2 (3)

Q3 (5143)

Q4 (576)

Q5 (80)

Q6 (100)

Q7 (7744)

Q8 (52)

Q9 (136)

Q1 (77)

FARMER (6)

A, E, F, M, R, R

A					
E					
F	A	E			
F	A	M			
F	A	R	E		
F	A	R	M	E	R

$$\begin{aligned}
 \frac{|5|}{|2|} - |4| &= 60 - 24 = 36 \\
 \frac{|3|}{|2|} - |2| &= 3 - 2 = 1 \\
 &= 1 \\
 &= 2 \\
 &= 1 \\
 \hline
 &77
 \end{aligned}$$

Q2 (3)

Total Number of Triangles = ${}^{15}C_3$

$i + j + k = 15$ (Given)

5 Cases			4 Cases			3 Cases			1 Cases		
i	j	k	i	j	k	i	j	k	i	j	k
1	2	12	2	3	10	3	4	8	4	5	6
1	3	11	2	4	9	3	5	7			
1	4	10	2	5	8						
1	5	9	2	6	7						
1	6	8									

Number of Possible triangles using the vertices P_i, P_j, P_k such that $i + j + k \neq 15$ is equal to

$${}^{15}C_3 - 12 = 443$$

Option (3)

Q3 (5143)

A = 4-digit numbers divisible by 3

$$A = 1002, 1005, \dots, 9999.$$

$$9999 = 1002 + (n-1)3$$

$$\Rightarrow (n-1)3 = 8997 \Rightarrow n = 3000$$

B = 4-digit numbers divisible by 7

$$B = 1001, 1008, \dots, 9996$$

$$\Rightarrow 9996 = 1001 + (n-1)7$$

$$\Rightarrow n = 1286$$

$$A \cap B = 1008, 1029, \dots, 9996$$

$$9996 = 1008 + (n-1)21$$

$$\Rightarrow n = 429$$

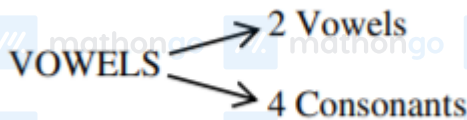
So, no divisible by either 3 or 7

$$= 3000 + 1286 - 429 = 3857$$

$$\text{total 4-digits numbers} = 9000$$

$$\text{required numbers} = 9000 - 3857 = 5143$$

Q4 (576)



All Consonants should not be together

$$= \text{Total} - \text{All consonants together,}$$

$$= 6! - 3!4! = 576$$

Q5 (80)

$$3n \text{ type} \rightarrow 3, 6, 9 = P$$

$$3n-1 \text{ type} \rightarrow 2, 5 = Q$$

$$3n-2 \text{ type} \rightarrow 1, 4 = R$$

$$\text{number of subset of } S \text{ containing one element which are not divisible by 3} = {}^2C_1 + {}^2C_1 = 4$$

number of subset of S containing two numbers whose sum is not divisible by 3

$$= {}^3C_1 \times {}^2C_1 + {}^3C_1 \times {}^2C_1 + {}^2C_2 + {}^2C_2 = 14$$

number of subsets containing 3 elements whose sum is not divisible by 3

$$= {}^3C_2 \times {}^4C_1 + ({}^2C_2 \times {}^2C_1) \times 2 + {}^3C_1 ({}^2C_2 + {}^2C_2) = 22$$

number of subsets containing 4 elements whose sum is not divisible by 3

$$= {}^3C_3 \times {}^4C_1 + {}^3C_2 ({}^2C_2 + {}^2C_2) + ({}^3C_1 {}^2C_1 \times {}^2C_2) \times 2$$

$$= 4 + 6 + 12 = 22.$$

number of subsets of S containing 5 elements whose sum is not divisible by 3.

$$= {}^3C_3 ({}^2C_2 + {}^2C_2) + ({}^3C_2 {}^2C_1 \times {}^2C_2) \times 2 = 2 + 12 = 14$$

number of subsets of S containing 6 elements whose sum is not divisible by 3 = 4

$$\Rightarrow \text{Total subsets of Set A whose sum of digits is not divisible by 3} = 4 + 14 + 22 + 22 + 14 + 4 = 80.$$

Q6 (100)

5	a	b	b	a	5
---	---	---	---	---	---

It is always divisible by 5 and 11.

$$\text{So, required number} = 10 \times 10 = 100$$

Q7 (7744)

209, 220, 231, ..., 495

$$\text{Sum} = \frac{27}{2} (209 + 495) = 9504$$

Number containing 1 at unit place

$$\begin{array}{r} \underline{2} \quad \underline{3} \quad \underline{1} \\ 3 \quad 4 \quad 1 \\ 4 \quad 5 \quad 1 \end{array}$$

Number containing 1 at 10th place

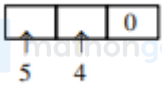
$$\begin{array}{r} \underline{3} \quad \underline{1} \quad \underline{9} \\ 3 \quad 1 \quad 9 \\ 4 \quad 1 \quad 8 \end{array}$$

$$\text{Required} = 9501 - (231 + 341 + 451 + 319 + 418)$$

$$7744$$

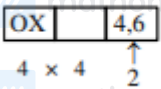
Q8 (52)

(i) When '0' is at unit place



Number of numbers = 20

(ii) When 4 or 6 are at unit place



Number of numbers = 32

So number of numbers = 52

Q9 (136)

$${}^1P_1 + 2 \cdot {}^2P_2 + 3 \cdot {}^3P_3 + \dots + 15 \cdot {}^{15}P_{15}$$

$$= 1! + 2 \cdot 2! + 3 \cdot 3! + \dots + 15 \times 15!$$

$$= \sum_{r=1}^{15} (r+1-1)r!$$

$$= \sum_{r=1}^{15} (r+1)! - (r)!$$

$$= 16! - 1$$

$$= {}^{16}P_{16} - 1$$

$$\Rightarrow q = r = 16, s = 1$$

$${}^{q+s}C_{r-s} = {}^{17}C_{15} = 136$$